

Nirav Diwan

Ph.D. student in C.S., University of Illinois at Urbana-Champaign

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Education

Aug 2024 –	University of Illinois Urbana-Champaign (UIUC) Ph.D. in Computer Science	Champaign, US
Aug 2022 Aug 2024	University of Illinois Urbana-Champaign (UIUC) M.S in Computer Science (thesis-track)	Champaign, US
Aug 2017 Jun 2021	Indraprastha Institute of Information Technology, Delhi (IIITD) Bachelor in Technology (B.Tech) in Computer Science & Engineering (CSE) 🏆 Dean's Award for Academic Excellence	Delhi, India

Publications

PR = In Preparation, S=In Submission, C=Conference, W=Workshop, P=Poster/Demo, J=Journal, *Equal Contribution

- [S.1] **CoT-Guard: Small Models for Strong Practical Monitoring** [🔗][📄]
Nirav Diwan*, Han Wang*, Berkcan Kapusuzoglu, Ramin Moradi, Giri Iyengar, Sambit Sahu, Huan Zhang, Gang Wang
The 40th Annual Conference on Neural Information Processing Systems, 2026 [NeurIPS 2026]
▶ We introduce a practical threat model for CoT Monitoring and train small models ($\leq 4B$) to outperform large models using RL.
- [W.3] **Extracting Sensitive Training Data from a Differentially Private Pre-trained LLM** [🔗]
Nirav Diwan, Gang Wang, Daniel Alabi
Theory and Practice of Differential Privacy (TPDP) Workshop, 2026 [TPDP 2026]
▶ We extract Personally Identifiable Information (PII) from VaultGemma — the first LLM trained through DP-SGD.
- [W.2] **Beyond BeautifulSoup: Benchmarking LLM-Powered Web Scraping for Everyday Users** [🔗]
Arth Bhardwaj, Nirav Diwan, Gang Wang
2nd Artificial Intelligence for Cyber Security (AICS) Workshop (Oral) @ AAAI, 2026 [AICS @ AAAI 2026]
▶ My awesome mentee (Arth) developed a benchmark to check if LLMs can scrape websites with minimal prompting!
- [C.6] **PurpCode: Reasoning for Safer Code Generation** [🔗][📄]
Jiawei Liu*, Nirav Diwan*, Zhe Wang*, Haoyu Zhai, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, Muntasir Wahed, Yinlin Deng, Hadjer Benkraouda, Yuxiang Wei, Lingming Zhang, Ismini Lourentzou, Gang Wang
39th The Annual Conference on Neural Information Processing Systems [NeurIPS 2025]
🏆 1st place in the Amazon Nova AI Challenge (\$250,000)
▶ Developed a reasoning model for secure code generation using Deliberative Alignment.
- [W.1] **Clear Preferences Leave Traces: Reference Model-Guided Sampling for Preference Learning** [🔗]
Nirav Diwan, Tolga Ergen, Dongsub Shim, Honglak Lee
1st Workshop on Preparing Good Data for Generative AI Challenges and Approaches, 2025 [Good Data Workshop @ AAAI 2025]
▶ Fine-tuned models identify quality preference pairs for alignment without knowing which response is better.
- [C.5] **MOCHA: Are Code Language Models Robust Against Multi-Turn Malicious Coding Prompts?** [🔗]
Muntasir Wahed, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, Nirav Diwan, Gang Wang, Dilek Hakkani-Tür, Ismini Lourentzou
30th Empirical Methods in Natural Language Processing (Findings), 2025 [EMNLP 2025]
▶ A multi-turn benchmark for jailbreaking code models
- [C.4] **You Can't Judge a Binary by Its Header: Data-Code Separation for Non-Standard ARM Binaries using Pseudo Labels** [🔗]
Hadjer Benkraouda, Nirav Diwan, Gang Wang
46th IEEE Symposium on Security and Privacy, 2025 [IEEE S&P 2025]
▶ We propose a novel way to separate data and code in non-standard ARM binaries using pseudo labels.
- [C.3] **It Doesn't Look Like Anything to Me: Using Diffusion Models to Subvert Visual Phishing Detectors** [🔗][📄]
Qingying Hao, Nirav Diwan, Ying Yuan, Mauro Conti, Giovanni Apruzzese, Gang Wang
33rd USENIX Security Symposium, 2024 [USENIX Security 2024]
▶ Used diffusion models to attack phishing detectors, validated across 100+ brands in white-box/black-box settings.
- [C.2] **Weakening the Inner Strength: Spotting Core Collusive Users in YouTube Blackmarket Networks** [🔗][📄]
Hridoy Sankar Dutta*, Nirav Diwan*, Tanmoy Chakraborty
16th International AAAI Conference on Web and Social Media, 2022 [ICWSM 2022]
▶ Investigated the collusive blackmarket on YouTube using graphs to identify the most influential blackmarket users.

[C.1] **Fingerprinting Fine-tuned Language Models in the Wild** [🔗] [📄]
Nirav Diwan, Tanmoy Chakraborty, Zubair Shafiq
59th Annual Meeting of the Association for Computational Linguistics (Findings), 2021 [ACL 2021]
🏆 **Dean's Thesis Appreciation Award**
➤ Developed an LLM-based classifier to attribute AI-generated text to its fine-tuned language model across 100+ classes.

[J.1] **RecipeDB: A Resource for Exploring Recipes** [🔗] [📄]
Devansh Batra*, Nirav Diwan*, Utkarsh Upadhyay*, Jushaan Singh Kalra*, Tript Sharma*, Aman Kumar Sharma*, Dheeraj Khanna*, Jaspreet Singh Marwah*, Srilakshmi Kalathil*, Navjot Singh*, Rudraksh Tuwani*, Ganesh Bagler*
Database: The Journal of Biological Databases and Curation, Oxford University Press, 2020 [Database 2020]
🌐 **Press:** [Times of India](#), [The National](#), [Nature India](#)
➤ Developed a worldwide database of recipes with over 100,000 recipes from 50+ countries.

Awards

Amazon Nova AI Challenge (Winner): Won 1st place as team co-lead in the [Amazon Nova AI Challenge](#) winning a cash prize of \$250K.

Amazon Nova AI Grant: Awarded grants worth \$250,000 + \$1.5 Million AWS Credits.

Catalyzing Advocacy in Science and Engineering (CASE) Workshop hosted by the AAAS (American Association for the Advancement of Science) 2024: Selected to represent the CS department from UIUC. Informed legislators about AI progress, and regulating AI with security and safety in mind.

Dean's Thesis Appreciation Award, IIITD: For outstanding thesis research from Dean of Academic Affairs.

Dean's Award for Academic Excellence, IIITD: For exceptional academic performance for the academic years 2019-20.

IIT-JEE 2017: Ranked 1884/1.5 million (top 0.15%) in IIT-JEE Mains and 4186/200,000 (top 2%) in IIT-JEE Advanced.

Recent Industrial Experience

1. **LG AI Research | Bilingual LLM Team** [🌐] Remote, US
Applied Science Intern | Area: Natural Language Processing (NLP) 2024
➤ Improved alignment of LLMs for single-turn and multi-turn settings. Results published in the **AAAI Good Data Workshop**.
➤ Developed a sampling method to be used on top of alignment methods (DPO, ORPO, SimPO) for Coding, Math, and Reasoning – increasing MT-Bench scores by +1.0 on average.
2. **Ema Inc. | Generative AI Team** [🌐] Remote, US
Applied Science Intern | Area: Natural Language Processing (NLP) 2023
➤ Led the large-scale deployment of a conversational AI agent for translating natural language queries into SQL (Text2SQL).
➤ Built an eval pipeline and benchmark dataset of 3,000+ real-world Text2SQL examples for robust evaluation.

Selected Research Experience

1. **University of Illinois Urbana-Champaign (UIUC) | Security & Privacy Group** [🌐] Champaign, US
Graduate Researcher | Advisor: [Prof. Gang Wang](#) | Areas: Machine Learning (ML) and Security & Privacy (S&P) 2022 - Present
➤ Identifying and protecting against practical underperformance, attacks, and misuse of AI models.
➤ Published projects at **NeurIPS**, **IEEE S&P** and **USENIX Security**.
2. **IIIT Delhi | Laboratory for Computational Social Systems (LCS2)** [🌐] Delhi, India
Undergraduate Researcher | Advisor(s): [Prof. Zubair Shafiq](#), [Prof. Tanmoy Chakraborty](#) | Areas: NLP, S&P and Graphs 2019 - 2021
➤ Identified malicious actors online using machine learning techniques.
➤ Published projects at **ACL** and **ICWSM**.

Teaching Experience

1. **Introduction to Computer Science (CS124) | UIUC** [🌐] Champaign, US
Teaching Assistant | Professor: [Prof. Geoffrey Challen](#) Fall 2022, Spring 2023, Fall 2023, Fall 2025
➤ Served as Teaching Assistant for three consecutive semesters, engaging with 1,300+ students in each semester.
➤ Developed course materials, supervised quizzes, and held office hours.
2. **Machine Learning Graduate (CS563) | IIIT Delhi** Delhi, India
Teaching Assistant | Professor: [Prof. Tanmoy Chakraborty](#) Spring 2020
➤ Assisted teaching Graduate Machine Learning course to 180 students across undergraduate, master's, and doctoral levels.
➤ Conducted tutorials, resolved queries, graded quizzes, and developed the mid-semester examination.

Talks

Can LLMs identify Security Vulnerabilities? <i>Advanced Computer Security, UIUC</i> Slides	April 2025
Jailbreaking of LLMs <i>LLM Post Pre-training, UIUC</i> Slides	October 2024
Generating Sequences by Learning to Self-Correct <i>Conversational AI, UIUC</i> Slides	May 2024
Promptly Racing Ahead: A Survey on Multi-task Prompt-Based Learning <i>Transfer Learning, UIUC</i> Slides	May 2023
Editing Models with Task Arithmetic <i>Transfer Learning, UIUC</i> Slides	April 2023
A pseudo-labelling approach for low-resource NLP Domain Adaptation <i>Trustworthy Machine Learning, UIUC</i> Slides	December 2022
Weight Poisoning Attack on Pre-trained Language Model <i>Security Machine Learning Seminar</i> Slides	July 2021
Practical No-box Adversarial Attacks against DNNs <i>Security Machine Learning Seminar</i> Slides	June 2021

Academic Service

Reviewer	NeurIPS 2026, TGAIR ICML 2026, ACM COMPASS’21, ICLR’22 (Student Reviewer)
Volunteer	ACM FAccT’23, AAAI ICSWM’21, AAAI ICWSM’22, ACL’21